

Western University
Department of Mechanical & Materials Engineering

MME 2273B – INTRODUCTION TO FLUID MECHANICS AND HEAT TRANSFER

COURSE OUTLINE – 2025-2026

***CALENDAR
DESCRIPTION:***

An introduction to fluid mechanics and heat transfer. The fluid mechanics part covers fluid properties, fluid statics including buoyancy and stability, one-dimensional fluid dynamics including conservation of mass and energy and losses in pipe networks. Heat transfer covers development of the general energy equation for three dimensions and steady-state conduction in one and two dimensions.

***COURSE
INFORMATION:***

Instructor: Dr. Rex Sherratt
Email: asherra@uwo.ca
Lectures/tutorials/labs: See [Draft My Schedule](#)

***CONSULTATION
HOURS:***

By appointment

PREREQUISITES:

NMM 2270A/B or the former Applied Mathematics 2270A/B

ANTIREQUISITES:

MSE 2273A/B

***ACCREDITATION
UNITS:***

Science = 50%, Engineering Science = 50%.

TOPICS:

- Fluid Statics
- Fluid Dynamics
- Introduction to heat transfer modes
- Steady state conduction

**GENERAL
LEARNING
OBJECTIVES (CEAB
GRADUATE
ATTRIBUTES)**

Knowledge Base (KB)	I	Use of Engineering Tools		Impact on Society and the Environment	
Problem Analysis (PA)	I	Individual and Teamwork	I	Ethics and Equity	
Investigation (I)	I	Communication Skills		Economics and Project Management	
Design		Professionalism		Life-Long Learning	

Notation: where x be I: Introductory, D: Intermediate, A: Advanced, or empty. I – The instructor will introduce the topic at the level required. It is not necessary for the student to have seen the material before. D – There may be a reminder or review, but the student is expected to have seen and been tested on the material before taking the course. A – It is expected that the student can apply the knowledge without prompting (e.g. no review).

**LEARNING
OUTCOMES:**

The Mechanical and Materials Engineering Program has been accredited by Canadian Engineering Accreditation Board (CEAB) of Engineers Canada. Accredited programs provide the academic requirements for licensure as a professional engineer in Canada. Western Engineering has defined indicators of the 12 Graduate Attributes (GAs) that the CEAB expects graduating engineering students to demonstrate. The connections between course learning outcomes and [Western Engineering's GA Indicators](#) are identified below.

Upon successful completion of this course, students will be able to:

1. Fluid Statics

- Identify the fluid properties required to solve a problem and determined their magnitude (KB2)
- Determine the pressure at a point in a static fluid from a measurement taken at another point (KB2, PA2)
- Quantify hydrostatic forces and moments acting on a system (KB3, PA2)
- Determine whether a partially submerged body will float and whether it will be stable (KB2, PA2)

2. Fluid Dynamics

- Apply the mass and energy conservation (Bernoulli) equations to solve idealized fluid flow problems (KB2, KB3)
- Determine flow rates and or pressure losses in a realistic pipe system (KB2, KB3, PA2)
- Successfully conduct and report laboratory measurements taken on a pipe flow system (I2, I3)

3. Introduction to Heat Transfer Modes

- Identify the modes of heat transfer associated with a given heat transfer problem (KB2)
- Calculate heat transfer rates and/or material/fluid temperatures for simple problems involving two or three heat transfer modes (KB3)

4. Steady state conduction

- a. Apply the heat diffusion equation to solve 1-D conduction problems in Cartesian, cylindrical, and spherical co-ordinate systems (KB3, PA2)
- b. Successfully conduct and report on laboratory measurements of linear conduction heat transfer (I2, I3)
- c. Determine the heat transfer characteristics of a finned surface
- d. Use shape factors to solve simple multi-dimensional heat transfer problems (KB3, PA2)

CONTACT HOURS: 3 lecture hours, 2 tutorial hours, 0.5 laboratory hours per week (laboratory hours occur in two 3-hour sessions), 0.5 course.

TEXTBOOK: **For Fluid Mechanics:** *Fluid Mechanics*, White F and Xue H, 9th Edition, McGraw Hill
For Heat Transfer: *Fundamentals of Heat and Mass Transfer*, Bergman T L, Lavine A S, 8th Edition, John Wiley and Sons

Both textbooks for this course (and alternative formats) can be found using the following link:

https://bookstore.uwo.ca/textbook-search?campus=UWO&term=W2025B&courses%5B0%5D=001_UW/MME2273B

Please note: Students may use different editions or copies of the textbook(s). As problem numbering can vary between editions, it is the students' responsibility to identify the equivalent problem in their copy.

UNITS: SI will be used; however, English units may be introduced through examples as required.

EVALUATION: **Overall:**

In-tutorial Questions	8% (1.6% each)
Quizzes/midterms	27% (13.50% each)
Laboratory	5% (2.5% each)
Final Examination	60%

Tentative Schedule:

In-tutorial Questions:

In-tutorial Question #1: Thursday 22nd January 2026
 In-tutorial Question #2: Thursday 29th January 2026
 In-tutorial Question #3: Thursday 13th February 2026
 In-tutorial Question #4: Thursday 26th February 2026
 In-tutorial Question #5: Thursday 12th March 2026
 In-tutorial Question #6: Thursday 19th March 2026
 In-tutorial Question #7: Thursday 26th March 2026

Quizzes:

Quiz #1 (Designated Assessment): Thursday 5th February 2025 (week 5), 1:40 pm – 3:10 pm
 Quiz #2: Thursday 5th March 2025 (week 8), 1:40 pm – 3:10 pm

Laboratories:

Laboratory experiment 1: Losses in pipe networks (Weeks 10 – 11)
 Laboratory experiment 2: Linear heat conduction (Weeks 11 – 12)

Final Examination:

The final examination will take place during the regular examination period.

COURSE POLICIES: The following course-specific policies will be strictly enforced throughout the course:

General: Students are responsible for regularly checking their Western e-mail and the Brightspace course website (<https://wts.uwo.ca/owl/>) in order to make themselves aware of any information that is posted about the course. If a student fails to act on information that has been posted on the Brightspace course website and does so without a legitimate explanation (i.e. those covered under the illness/compassionate form), then there are no grounds for an appeal.

Quizzes (midterms): Both Quizzes will be closed-book and an equation sheet will be provided in the exam. Only non-programmable calculators will be allowed. Students arriving more than 30 minutes late for an in-person Quiz will not be allowed to write the Quiz and will receive zero marks. If a student is excused from writing a Quiz by academic consideration (e.g. due to illness), the weighting of that Quiz will be placed onto the Final Exam. If a student is going to miss, or require accommodations for, a Quiz for religious reasons, they must inform the instructor in writing within 1 week of the start of the term or they will be required to write the Quiz. Even with academic consideration (see above) a student must attend and submit work for at least 1 of the 2 Quizzes to meet the minimum coursework requirements (see Summary of Minimum Coursework Requirements Policies below). **The first quiz is Designated Assessment for this course.** There will be no makeup quizzes.

Weekly In-tutorial Questions: The in-tutorial questions are open-book and students must work in a group of 4 students, submitting a single solution for marking with all the group members' names and ID numbers on it. Individual students must also submit their own work on the problem; failure to complete both steps may result in a grade of zero. Students arriving more than 30 minutes late for an in-tutorial question will not be allowed to be included on a submitted solution and will receive zero marks. Even with any academic considerations (see above), a student must attend and submit work for a minimum of 5 out of the 7 in-tutorial questions to meet the minimum coursework requirements (see Summary of Minimum Coursework Requirements Policies below). The total course grade for in-tutorial questions is based on the student's best five submissions.

Laboratories: Laboratory reports are to be submitted during the same laboratory class in which the experiment is completed. Attendance at both of the designated laboratory sessions is compulsory (even with academic consideration). Failure to attend and complete both laboratory sessions will mean that the student will not meet the minimum coursework requirements (see Summary of Minimum Coursework Requirements Policies below). If a laboratory is missed without academic consideration, there is no guarantee of a make-up laboratory session. Under extenuating circumstances, as determined by the instructor, an alternative lab format may be used. If at-home labs are assigned with a week or more between the date they are assigned and the date they are due, students who receive academic consideration for the date the lab is due must submit their lab within forty-eight hours of the original deadline. Late labs will not be accepted.

Final Exam: The final exam will be closed-book and an equation sheet will be provided. Only non-programmable calculators will be allowed. If a minimum mark of 50% is not obtained on the final exam the student cannot receive a final course mark greater than 48%. Students arriving more than 30 minutes late for an in-person exam will not be allowed to write the exam and will receive zero marks.

Summary of Minimum Coursework Requirements Policies: Even taking into account any academic considerations, a student must:

- (1) Attend and submit solutions for at least 5 of the 7 In-tutorial questions;
- (2) Attend and submit work for at least 1 of the 2 Quizzes and
- (3) Attend and submit work for both of the laboratory sessions.

Any student not meeting these requirements may not receive a grade higher than 48% in the course. In the absence of academic consideration, it is necessary to submit work for 5 in-tutorial questions and sit both quizzes to have the opportunity to gain full marks in those categories. Late assessments will not be accepted, and the student will receive a zero on assessments that are not submitted unless the student has academic consideration. There will be no makeup quizzes.

Attendance: Any student who, in the opinion of the instructor, is absent too frequently from class, laboratory, or tutorial periods will be reported to the Dean (after due warning has been given). On the recommendation of the department, and with the permission of the Dean, the student will be debarred from taking the regular final examination in the course.

Students are required to behave professionally during all class meetings. Failure to comply may result in the student receiving a grade of zero on any associated grades or assessments occurring at the time of the infraction.

Use of Electronic Devices: Only a non-programmable calculator may be used in the Quizzes and in the Final Exam.

Policy on Repeating All Components of a Course: Students who are required to repeat an Engineering course must repeat all components of the course. No special permissions will be granted enabling a student to retain laboratory, assignment, or test marks from previous years. Previously completed assignments and laboratories cannot be resubmitted by the student for grading in subsequent years.

General Faculty / University Policies

In the event of contradictions between course-specific policies above and general Faculty / University policies described below, please contact your course instructor for clarification.

Missed/Late Accommodation Policy

1. Students missing a test/assignment/lab or examination you will report the absence by submitting an Academic Consideration Request form through [STUDENT ABSENCE PORTAL](#).
2. **Documentation must be provided as soon as possible.**

Exam Accommodation

1. If you are unable to write a final examination, report your absence using the Academic Consideration Request Form through [STUDENT ABSENCE PORTAL](#).
2. Be prepared to provide the Undergraduate Services Office with supporting documentation (below for information on documentation) the next day, or as soon as possible (in cases where students are hospitalized). The following circumstances are not considered grounds for missing a final examination or requesting special examinations: common cold, headache, sleeping in, misreading timetable and travel arrangements.
3. In order to receive permission to write a Special Examination, you must obtain the approval of the Chair of the Department and the Associate Dean and in order to apply you must submit an Academic Consideration Request Form through [STUDENT ABSENCE PORTAL](#).

PLEASE NOTE: It is the student's responsibility to check the date, time and location of the Special Examination.

Late Assignments

1. Advise the instructor if you are having problems completing the assignment on time (prior to the due date of the assignment).
2. Be prepared to submit the Academic Consideration Request Form and provide documentation if requested by the instructor (see below for information on documentation).
3. If you are granted an extension, establish a due date. The approval of the Chair of your Department (or the Assistant Dean, First Year Studies, if you are in first year) is not required if assignments will be completed prior to the last day of classes.
4. Some courses may have built-in flexibility for assignment deadlines or the total number of assignments that will be graded. See course-specific policies for details.
5. Extensions beyond the end of classes must have the consent of the instructor, the department Chair and the Associate Dean, Undergraduate Studies. Documentation is mandatory.

Note: Forged notes and certificates will be dealt with severely. To submit a forged document is a scholastic offence (see below).

Medical Accommodation

1. The Academic Consideration Request Form is available through the [STUDENT ABSENCE PORTAL](#).
2. Requests for academic consideration must include the following components:
 - a. Indication of the course(s) and assessment(s) affected by the request
 - b. Medical note, and
 - c. Additional supporting documentation as relevant
3. Requests for academic consideration without a medical note or other supporting documentation may be accepted once per term, per course.

4. Undocumented absences cannot be used for examinations scheduled by the Office of the Registrar during official examination periods (including take-home final exams and December mid-year exams for full courses) and practical laboratory and performance tests typically scheduled in the last week of the term. Undocumented absences also cannot be used for the “designated assessment” in each course. When flexibility in assessment exists and is clearly stated on the course outline, both undocumented absences and academic consideration requests with documentation may be denied.
5. **Students must request academic consideration as soon as possible and no later than 48 hours after the missed assessment.**
6. Once the request and supporting documents have been received and reviewed, appropriate academic consideration, if granted, shall be determined by the instructor in consultation with the academic advisor, in a manner consistent with the course outline.

Academic consideration may include extension of deadlines, waiver of attendance requirements for classes/labs/tutorials, or re-weighting of course requirements. Some forms of academic consideration, such as arranging Special Examinations, assigning a grade of Incomplete, or granting late withdrawals without academic penalty, may only be granted by the Academic Advising office of the Faculty of Registration.

7. An instructor may deny academic consideration for any assessment that is not required in the calculation of the final grade (e.g., “8 of 10 quizzes”). Assessment flexibility must be indicated on the course outline.
8. An instructor may deny academic consideration relating to the timeframe submission of work where there is already flexibility in the submission timeframe (e.g., 72-hour submission window). This assessment flexibility must be indicated on the course outline.

Religious Accommodation

When scheduling unavoidably conflicts with religious holidays, which (a) require an absence from the University or (b) prohibit or require certain activities (i.e., activities that would make it impossible for the student to satisfy the academic requirements scheduled on the day(s) involved), no student will be penalized for absence because of religious reasons, and alternative means will be sought for satisfying the academic requirements involved. If a suitable arrangement cannot be worked out between the student and instructor involved, they should consult the appropriate Department Chair and, if necessary, the student's Dean.

It is the responsibility of such students to inform themselves concerning the work done in classes from which they are absent and to take appropriate action.

Academic Integrity

In the Faculty of Engineering, we encourage students to create a culture of honesty, trust, fairness, respect, responsibility, and courage, befitting the professional degree you are pursuing.

Please visit [Academic Integrity Western Engineering](#) for more information

Academic Offences

Plagiarism means using another's work without giving credit. The university has rules against plagiarism and other scholastic offences. Western Engineering has a zero-tolerance policy on plagiarism. The minimum penalty is zero on the course work and a repeat offence will earn you zero on the course. A third offence may lead to expulsion from the university.

[Scholastic Discipline for Undergraduate Students](#) & [Cheating, Plagiarism and Unauthorized Collaboration: What Students Need to Know](#)

Students must write their reports, essays and assignments in their own words. Whenever students take an idea or a passage from another author, they must acknowledge their debt both by using quotation marks where appropriate and by proper referencing such as footnotes or

citations. University policy states that cheating, including plagiarism, is a scholastic offence. The commission of a scholastic offence is attended by academic penalties, which might include expulsion from the program. If you are caught cheating, there will be no second warning.

All required papers may be subject to submission for textual similarity review to commercial plagiarism detection software under license to the University for the detection of plagiarism. All papers submitted will be included as source documents on the reference database for the purpose of detecting plagiarism of papers subsequently submitted to the system. Use of the service is subject to the licensing agreement, currently between the University of Western Ontario and Turnitin.com (<http://www.turnitin.com>). Scholastic offences are taken seriously and students are directed to read the appropriate policy, specifically, the definition of what constitutes a Scholastic Offence, in the relevant section of the Academic Handbook: http://www.uwo.ca/univsec/pdf/academic_policies/appeals/scholastic_discipline_undergrad.pdf

Faculty of Engineering AI Policy

The use of generative Artificial intelligence (GenAI) tools won't be discouraged in the Faculty of Engineering. As we pride ourselves on building the future we can't hide from the use of GenAI tools to contribute to the understanding of the course materials. However, the use of GenAI tools in any assignment or contribution during the course will have to be disclosed, as a resource.

GenAI tools use won't be permitted in any type of examination or other assessments where the faculty have prohibited their use. If use of GenAI tools is detected by the instructor in these instances, academic offences penalties might be imposed against the student.

Use of English Policy

In accordance with Senate and Faculty Policy, students may be penalized up to 10% of the marks on all assignments, tests, and examinations for improper use of English. Additionally, poorly written work except for the final examination may be returned without grading. If resubmission of the work is permitted, it may be graded with marks deducted for poor English and/or late submission.

Accessibility

Western is committed to achieving barrier free accessibility for persons with disabilities studying, visiting and working at Western. As part of this commitment, there are a variety of services, groups and committees on campus devoted to promoting accessibility and to ensuring that individuals have equitable access to services and facilities. To help provide the best experience to all members of the campus community, please visit the [Accessibility Western University](#) for information on accessibility-related resources available at Western.

Students with disabilities may arrange for academic accommodation at Western. For a more detailed explanation, please visit [Academic Support & Engagement -Academic Accommodation](#).

Inclusivity, Diversity, and Respect

The Faculty of Engineering at Western University is committed to creating equitable and inclusive learning environments that value diverse perspectives and experiences. We recognize that university courses often marginalize students based on social identity characteristics such as, but not limited to, Indigeneity, race, ethnicity, nationality, ability, gender identity, gender expression, sexuality, age, language, religion, and socioeconomic status. Understanding this, we strive to facilitate equitable experiences and inclusion within the classroom by respecting and integrating multiple ways of knowing, being, and doing. Please visit the [Office of Equity, Diversity and Inclusion](#).

Health and Well-Being

- [Health & Wellness Services – Students](#) - Offers appointment-based medical clinic for all registered part-time and full-time students.
- [Mental Health Support](#) - Provides professional and confidential services, free of charge, to students needing assistance to meet their personal, social and academic goals. Services include consultation, referral, groups and workshops, as well as brief, change-oriented psychotherapy.

- [Crisis Support](#) - For immediate assistance, please visit Thames Hall Room 2170 or call 519-661-3030. The crisis clinic operates between 11:00 am - 4:30 pm. For after-hours crisis support, click [here](#).
- [Gender-Based Violence and Survivor Support](#) - Western [is committed to reducing incidents of gender-based and sexual violence](#) and providing compassionate support to anyone who has gone through these traumatic events. If you have experienced gender-based or sexual violence (either recently or in the past), you will find information about support services for survivors, including emergency contacts, [here](#). To connect with a case manager or set up an appointment, please contact support@uwo.ca.

Important Links

- [WESTERN ACADEMIC CALENDAR](#)
- [ACADEMIC RIGHTS AND RESPONSIBILITIES](#)
- [ENGINEERING PROGRESSION REQUIREMENTS AND ACADEMIC REGULATIONS](#)
- [UNIVERSITY STUDENTS' COUNCIL \(USC\) - SERVICES](#)
- [IMPORTANT DATES AND DEADLINES](#)
- [ACADEMIC CONSIDERATION FOR MEDICAL ILLNESS - UNDERGRADUATE STUDENTS](#)
- [ACCOMMODATIONS FOR RELIGIOUS HOLIDAYS](#)
- [SCHEDULING OF ASSIGNMENTS, TESTS, AND EXAMINATIONS](#)
- [STUDENT FORMS](#)
- [OFFICE OF THE REGISTRAR](#)
- [RETENTION OF ELECTRONIC VERSION OF COURSE OUTLINES \(SYLLABI\)](#)
- [ACADEMIC APPEALS](#)
- [STUDENT ABSENCE PORTAL](#)

Note: These instructions apply to all students registered in the Faculty of Engineering regardless of whether the courses are offered by the Faculty of Engineering or other faculties in the University.

Add Deadlines:

First term half course (i.e. "A" or "F")	September 13, 2024
Full courses and full-year half course (i.e. "E", "Y" or no suffix)	September 13, 2024
Second term half course (i.e. "B" or "G")	January 14, 2025

Drop Deadlines:

First term half course without penalty (i.e. "A" or "F")	November 12, 2024
Full courses and full-year half courses without penalty (i.e. "E", "Y" or no suffix)	December 2, 2024
Second term half or second term full course without penalty (i.e. "B" or "G")	March 7, 2025

Contact Information:

Undergraduate Services Office:	SEB 2097
Phone: 519-661-2130	E-mail: engugrad@uwo.ca
Mechanical Engineering:	SEB 3002
Phone: 519-661-4122	E-mail: mmeundergraduate@uwo.ca
Chemical & Green Process Engineering:	TEB 477
Phone: 519-661-2131	E-mail: cbeugrad@uwo.ca
Civil Engineering:	SEB 3005
Phone: 519-661-2139	E-mail: civil@uwo.ca
Computer, Electrical, Mechatronic Systems & Software Engineering	TEB 279
Phone: 519-661-3758	E-mail: eceugrad@uwo.ca
Integrated Engineering	ACEB 2410
Phone: 519-661-6725	E-mail: engceli@uwo.ca
Office of the Registrar/Student Central	WSSB 1120
Phone: 519-661-2100	